

Experimental designs

Sources of Bias

Trial Ethics and historical
atrocities

Ethical principles and
current regulations

L07: Randomized designs, bias and ethics

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Learning objectives for today

- ▶ Name the main parts of a randomized trial
- ▶ Spot ways a study can still be biased after randomization
- ▶ Connect trial ethics to the rules used today

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Who (or what) receives the assigned treatment?

- ▶ **Individuals:** a patient gets Drug A or Drug B
- ▶ **Households:** a family gets a water filter or usual water
- ▶ **Groups:** a classroom gets a new curriculum or the old one
- ▶ **Clinics:** a clinic starts a reminder system or continues as usual

The unit often becomes a **row** in the data. If we measure the same unit over time, we may have several rows per unit (*long* / longitudinal data).

Discuss: pick one level and invent an exposure that fits it.

Randomization means treatment is assigned **by chance**, not by the researcher's preference.

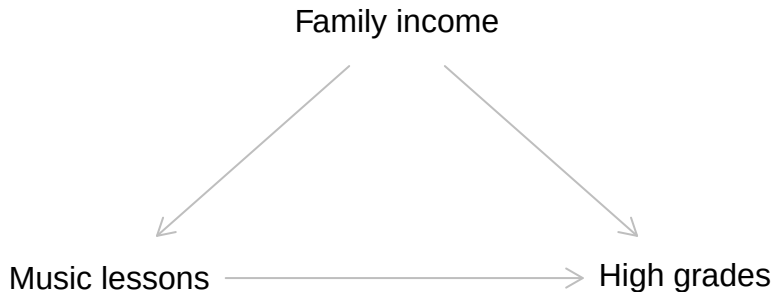
Why bother?

- ▶ People who choose (or are chosen for) treatment are often different at the start
- ▶ Chance tends to balance those starting differences across groups

What does randomization *not* fix by itself?

Music lessons and grades (no randomization)

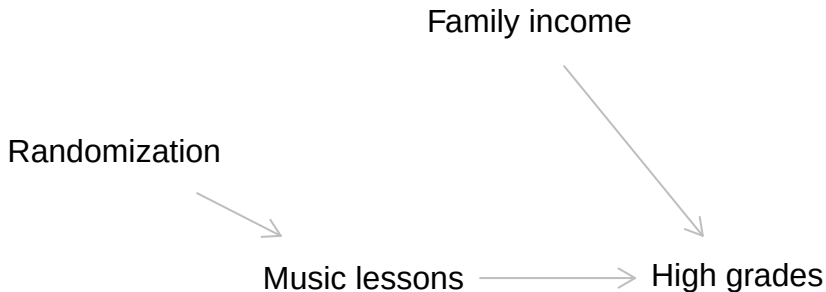
Setting: we watch students in ordinary life. Some take music lessons; some do not. We then look at grades.



Question: why might lesson-takers have better grades *even if* lessons do nothing?

Music lessons and grades (with randomization)

Setting: eligible students are assigned **by chance** to lessons or no lessons. Family income can still affect grades — but it no longer decides who gets lessons.



Compare the two DAGs: which arrow disappeared?

A simple randomized trial

A “clean” starter picture:

- ▶ units look a lot alike
- ▶ one treatment group and one control group
- ▶ simple chance assignment
- ▶ outcome shows up soon

Example: fever medicine vs placebo; temperature checked the same evening.

Real trials are messier — but this picture is the idea we stretch.

Every effect answer needs a **comparison**.

Ideal idea: a **counterfactual** — what would the treated group have looked like under the other exposure?

In a trial we recruit people (or clinics, etc.) and assign them by chance to:

- ▶ intervention, or
- ▶ control (usual care, older drug, placebo, ...)

Discuss: why not one treated person vs one control person?

Setting: 50,000 people all get a vaccine. No one is held back for comparison. In the next 3 weeks, 0.1% report an adverse event.

Does this show the vaccine is harmful?

What is missing?

- ▶ We do not know the **background rate** of the same events among similar people who did **not** get the vaccine

What is an RCT?

RCT = randomized controlled trial

- ▶ R — chance assigns treatment
- ▶ C — there is a planned comparison group
- ▶ T — it is an experiment under a protocol

If one letter is missing, it is not an RCT — even if the sample is huge.

Other common trial terms

- ▶ **Cluster randomized:** assign whole groups together (schools, clinics)
- ▶ **Community randomized:** assign whole communities
- ▶ **Blind / double / triple blind:** who does *not* know the assignment
- ▶ **Placebo controlled:** comparison arm uses an inactive look-alike

For each label, ask: who got what, and who knew?

Blinding

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Blinding = keep the treatment assignment hidden from someone.

Why? Knowing the arm can change what people report, look for, or score.

- ▶ **Single blind:** usually participants
- ▶ **Double blind:** participants + people scoring outcomes
- ▶ **Triple blind:** often also investigators / analysts

(Definitions vary — read the methods.)

When is full blinding hard? Surgery, exercise, counseling, cooking classes. . .

Then what can we still do?

Placebo

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A **placebo** looks like treatment but has no active ingredient.

It helps with blinding, and it helps separate drug effects from expectations and clinic care.

When is a pure placebo arm a problem?

What is the placebo effect?

People can feel better after inactive treatment. Reasons can include expectation, care, and time (recovery on its own).



[*NYT Magazine*](#), Nov 2018

Trial reading tip: both arms may improve. Look at the **difference**.

Why not always randomize?

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If RCTs are so strong, why not learn *only* from them?

- ▶ Who joins / who stays may be a selected group
- ▶ Cost, logistics, feasibility
- ▶ Strict trial conditions $\neq$ everyday life (**efficacy** vs **effectiveness**)
- ▶ Some exposures cannot be assigned (or should not be)
- ▶ Answers can take longer than a policy needs

So: strong tool when it fits — not the only tool.

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Randomization balances arms **inside the enrolled sample**.

It does not fix a sampling frame that missed part of the target population (**under-coverage**).

Example: a depression drug trial recruits only from specialty clinics. Who never had a chance to enter?

Small samples are noisy (think: eight online reviews).

Rare events and small benefits need more people before chance is an unlikely explanation.

Size helps **precision**. It does not wash away **bias**.

Two different moments:

1. **Non-response:** invited people decline to join
2. **Lost to follow-up:** enrolled people disappear later

If one arm loses more people — especially sicker people — the comparison can bend after a fair start.

Look at the study flowchart: screened → randomized → analyzed.

Contamination: people in the “wrong” arm get (enough of) the treatment.

Example setting (lifestyle): control participants hear the study goals, then change diet/exercise on their own. Arms look more alike. The estimated effect can shrink toward **no difference**.

Related cases: **MRFIT** (heart risk counseling) and **SHAZ!** (community HIV program).

Setting: 200 people are randomized. Some stop the pills early.

Two common analyses:

- ▶ **Intention to treat (ITT):** keep people in the arm they were **assigned**
 - ▶ answers: “what happened when we *offered* this plan?”
- ▶ **Per protocol:** analyze by what people **actually took**
 - ▶ can describe adherers — but adherers are a selected group

If ITT and per protocol disagree, ask why.

Randomization does not fix bad measurement.

- ▶ imprecise tools
- ▶ confusing questions
- ▶ the measuring situation itself
 - ▶ **white-coat hypertension**: BP rises in clinic
 - ▶ **sleep lab**: sleep may change just because you are in a lab

Random error adds noise. **Systematic error** pushes results in one direction — and big n will not cancel it.

People may not report (or remember) the full story:

- ▶ **social desirability**: answers that sound “good”
- ▶ **poor recall**: forgetting
- ▶ **biased recall**: a bad outcome changes how people search memory
- ▶ **diary non-compliance**: filled in later from a blurry week

Ask: is the primary outcome self-report, and could one arm feel more judged?

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Trial Ethics and historical atrocities

This is a methods course — and the “units” are often people.

Next: a few historical cases that help explain today’s rules (consent, IRBs, equipoise).

If you need to step away, please do.

Is randomization ethical (when it is possible)?

Possible $\neq$ okay to do.

Is it ethical to randomize people to:

- ▶ lack of sleep?
- ▶ exposure to an infectious disease?
- ▶ fetal exposure via maternal smoking or alcohol?

What would you need to know before saying yes?

Tuskegee syphilis study

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Setting (1932–1972, Macon County, Alabama): U.S. Public Health Service followed hundreds of Black men with syphilis to watch the “natural” course of disease.

Men were told they were being treated. Effective care was withheld. Penicillin became standard around 1945 — and was still not provided. Planned for months; ran ~40 years; stopped after press exposure.

Ethical failures at Tuskegee

- ▶ No real **informed consent**
- ▶ Effective treatment withheld (no **equipoise**)
- ▶ Racism shaped who was studied: would the same design have been used with White men?

Timeline: **CDC**

Background: residential schools in Canada

Setting: state-supported schools (roughly 1879–1979) that removed Indigenous children from families.

- ▶ *150,000 children; ~80,000 survivors still alive*
 - ▶ languages/cultures forbidden; widespread abuse
 - ▶ early reports: very high mortality soon after entry
 - ▶ Truth and Reconciliation Commission: ~6,000 deaths (~1 in 25)

This is the world in which the nutrition experiments happened.

Nutrition experiments in residential schools

Setting: researchers studied malnutrition inside those schools.

- ▶ children were already hungry
- ▶ “control” and “treatment” groups were denied adequate food
- ▶ treatments were weak / sometimes harmful
- ▶ dental care was withheld so decay could be *observed*

What does “control” mean when the people cannot leave or refuse?

Ethical failures in the nutrition experiments

- ▶ no informed consent (parents not told; children could not consent)
- ▶ effective care withheld
- ▶ compulsory schooling later judged cultural genocide (Truth and Reconciliation Commission)

MacDonald, Stanwick & Lynk: nutrition research on residential school children and ethics in Indigenous health research.

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Key influential documents: Nuremberg

After WWII, the Nuremberg Code answered medical crimes by Nazi doctors.

It feeds into U.S. rules (Title 45/46 CFR) and IRB review. Human subjects research in the U.S. generally needs IRB approval first.

Nuremberg principles (1–3)

Ask as questions:

1. Was consent truly voluntary?
2. Is the knowledge worth human risk — and hard to get another way?
3. Does prior evidence justify moving to people now?

Nuremberg principles (4–7)

4. Avoid unnecessary suffering
5. Do not plan studies expected to kill or disable
6. Risk must fit the importance of the problem
7. Have real plans/facilities to protect people from harm

Nuremberg principles (8–10)

- 8. Qualified people run the work carefully
- 9. Participants can stop if continuing feels impossible
- 10. Leaders must stop the study if serious harm becomes likely

Consent is not only day 1. Monitoring and stopping rules matter later too.

Key influential documents: Belmont

The Belmont Report (1978) followed scandals including Tuskegee.

Three lenses used in U.S. review:

- ▶ respect for persons (autonomy + protection)
- ▶ beneficence (help more / harm less)
- ▶ justice (fair sharing of risks and benefits)

Consent needs more than a signature.

- ▶ What must people know? (purpose, risks, benefits, alternatives, right to leave)
- ▶ Who can consent? (adults; parents/guardians for children + assent when possible)

Can the person **understand** the form — and freely say no?

Even with a signed form, “yes” can be bent:

- ▶ payments that are hard to refuse
- ▶ study as the only path to scarce care
- ▶ fear that saying no hurts ordinary medical care

Useful test: *given this person's situation, could they decline?*

Equipoise: experts are genuinely unsure which arm is better.

- ▶ at the start, neither arm should be clearly known to be worse
- ▶ if one arm is already believed worse/more harmful — lottery is hard to justify
- ▶ early stopping: when uncertainty collapses, continuing can become unethical

TCPS 2 reference

Who benefits, who is included?

Justice asks:

- ▶ Do participants get any direct benefit?
- ▶ Could people *like* them benefit from the findings?
- ▶ Who is unfairly left out of the evidence base (e.g., pregnant people, older adults, common comorbidities)?

Data sovereignty: who owns the data?

Ethics continues after the last visit.

For Indigenous communities, data have often been collected, stored, and interpreted by outside institutions.

Ask: who owns the file? who approves reuse? who tells the story?

- ▶ FNIGC (2019); Kukutai & Taylor (eds., 2016)

Principles of Indigenous data sovereignty

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- ▶ extraction has reinforced unequal power
- ▶ framing has emphasized deficit / stereotypes
- ▶ state data are often of little use to Nations
- ▶ governments assumed ownership by default
- ▶ Nation-to-Nation governance conversations have been thin

Source: BCFNDGI / Open North, *Decolonizing Data* (2017)

Parting humor



"Miss Peterson, may I go home? I can't assimilate any more data today."

Three questions to keep

1. What was randomized — and who remained in the analysis?
2. Could measurement, adherence, or dropout still bend the result?
3. Who consented, who benefited, and who controls the data story?